MA3077 Operational Research
Problem Sheet 1

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Policy on Generative AI use: you can use generative AI tools to revisit concepts and develop your solution. However, you must not use these tools to directly solve problems or copy their outputs.

Notation and methods: to answer the following questions, you must employ the methods presented during the lectures and the notation used in the slides. Failure to comply with these requirements will result in significant deduction of marks.

Problem 1 (10 marks).

1. **(2 marks)** Convert the following linear program into standard form:

$$\max 3x_1 + 5x_2 \qquad \text{subject to} \quad \begin{aligned} &2x_1 + x_2 \leqslant 10 \\ &x_1 + 3x_2 \geqslant 15 \\ &x_1 - x_2 = 5 \\ &x_1, x_2 \geqslant 0 \end{aligned}$$

2. **(5 marks)** Given the primal problem:

$$\max 2x_1 + 3x_2 \qquad \text{subject to} \quad \begin{aligned} &x_1 + x_2 \leqslant 4 \\ &2x_1 + x_2 \leqslant 5 \\ &x_1, x_2 \geqslant 0 \end{aligned}$$

   (a) Draw the feasible set, determine an optimal solution graphically, and indicate which constraints are active.

   (b) Formulate the dual problem by providing the Lagrangian function, the Lagrangian multipliers and the dual function.

   (c) Find the optimal value of the dual problem using the strong duality theorem.

   (d) Explain the difference between weak and strong duality.

3. **(3 marks)** Use MATLAB's `linprog` function to the following linear program:

$$\max 3x_1 + 2x_2 \qquad \text{subject to} \quad \begin{aligned} &x_1 + x_2 \leqslant 5 \\ &2x_1 + x_2 \leqslant 6 \\ &x_1, x_2 \geqslant 0 \end{aligned}$$

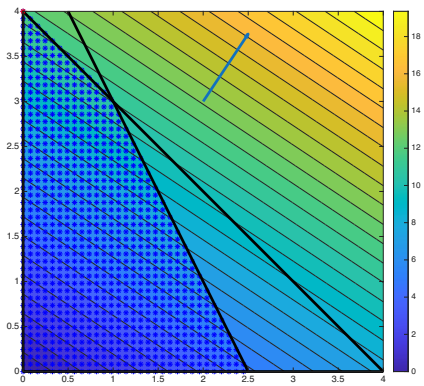
   You can either write the commands by hand or include your m-file in your submission.

   Hint: Convert the problem to minimisation form before using linprog.

Solution 1. 1. Introduce slack variables to obtain the standard form:

$$
\begin{aligned}
-\min -(3x_1 + 5x_2) \quad \text{subject to} \quad & 2x_1 + x_2 + s_1 = 10 \\
& x_1 + 3x_2 - s_2 = 15 \\
& x_1 - x_2 = 5 \\
& x_1, x_2, s_1, s_2 \geqslant 0
\end{aligned}
$$

2. (a) From the graphics



we see that the active constraints are $x_1 + x_2 = 4, x_1 = 0$. The optimal solution is $x^* = (0, 4)^T$.

(b) The primal problem is $\max c^T x$ subject to $Ax \leqslant b, x \geqslant 0$ where

$$c = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$$

Hence $L : \mathbb{R}^2 \times \mathbb{R}^2_+ \times \mathbb{R}^2_+$,

$$L(x, y, s) = c^T x + y^T(b - Ax) + s^T x = x^T(c - A^T y + s + y^T b$$

Then the dual function is given by

$$g(x, y) = \max_{x \in \mathbb{R}^2} L(x, y, s) = \begin{cases} y^T b & c - A^T y + s = 0 \\ \infty & \text{else} \end{cases}$$

The dual problem is hence

$$\min y^T b, \text{ s.t } \quad c - A^T y + s = 0, y, s \geqslant 0$$

which is equivalent to

$$\min y^T b, \text{s.t.} \quad c \leqslant A^T y, y \geqslant 0\,.$$

This can also be written as

$$
\begin{aligned}
\min\ 4y_1 + 5y_2 \quad \text{subject to:} \ & y_1 + 2y_2 \geqslant 2 \\
& y_1 + y_2 \geqslant 3 \\
& y_1, y_2 \geqslant 0
\end{aligned}
$$

(c) Since the primal solution has the finite optimal value $p^* = c^T \begin{pmatrix} 0 \\ 4 \end{pmatrix} = 12$ by the strong duality theorem, the dual problem has optimal value $d^* = p^* = 12$.

(d) Weak duality states that the optimal values p^* and d^* of the primal and dual problem satisfy $d^* \leqslant p^*$. If additionally at least one of the optimal values is finite, then so is the other and equality holds.

3. Since MATLABs linprog finds minimisers, we convert the problem to the minimum $\min(-3x_1 - 2x_2)$. Note that the optimal value of the problem d^* is then the negative of the optimal value of the converted problem:

```
f = [-3;-2];
Aeq = [];
beq = [];
A = [1,1; 2,1];
b = [5;6];
lb = zeros(size(f));
ub = [];

[x, fval] = linprog(f,A,b,Aeq,beq,lb,ub);
opt=-fval;
fprintf('The optimal controls are (x, y) = (%1.2f, %1.2f).N', x)
fprintf('The optimal value is p* = %1.2f.N', opt)
```

results in

```
The optimal controls are (x, y) = (1.00, 4.00).
The optimal value is p* = 11.00.
```

Problem 2 (10 points).

1. **(5 marks)** Solve the following linear program using the simplex method with $B = \{2, 4\}$:

$$\min 3x_1 + 2x_2 \qquad \text{subject to} \quad \begin{aligned} x_1 + x_2 + x_3 &= 4 \\ 2x_1 + x_2 - x_4 &= 2 \\ x_1, x_2, x_3, x_4 &\geqslant 0 \end{aligned}$$

2. **(5 marks)** Formulate the following optimisation problem as a linear programming problem.

$$\max x_1 + x_2 \qquad \text{subject to} \quad ||x||_1^2 \leqslant 9, \quad x = (x_1, x_2)^T \in \mathbb{R}^2 .$$

Solution 2. 1. The linear programming problem is already in standard form. We introduce the notation

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & -1 \end{pmatrix}, b = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, c^T = (3, 2, 0, 0).$$

We begin with the index set $B = \{2, 4\}$, which implies that

$$A_B = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, A_N = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \quad {c_B}^T = (2, 0),\ {c_N}^T = (3, 0).$$

It's useful to compute $A_B^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$. Then we compute the "modified objective"

$$c_N^T - c_B^T A_B^{-1} A_N = (2, 0) - (3, 1) \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} = (1, -2)$$

and select the second index $N(2) = 3$. To see how much we can increase $x_{N(2)}$, we compute

$$A_B^{-1} b = x_B = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

and

$$d = \max\{A_B^{-1} A_{N(1)}, 0\} = \max\{\begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, 0\} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and set$x_{N(2)} = \min \frac{(A_B^{-1} b)}{d} = \min\{\frac{4}{1}, \frac{2}{1}\} = 2$. To find out which index we should remove from B, we compute with the new $x_N = (0, 2)^T$

$$x_B = A_B^{-1}(b - A_N x_N) = \begin{pmatrix} 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

This implies that we need to replace $B(2)$ with $N(2)$. The new index set is $B = \{2, 3\}$, and

$$A_B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_N = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}, \quad c_B^T = (2, 0), c_N^T = (3, 0).$$

We compute again $A_B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$ and the new "modified objective"

$$c_N^T - c_B^T A_B^{-1} A_N = (3, 0) - (2, 0) \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = (-1, 2),$$

which implies that we now need to select the index $N(1) = 1$ to get a decrease in the objective value. To see how much we can increase $x_{N(1)}$, we compute

$$A_B^{-1} b = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

and

$$d = \max\{A_B^{-1} A_{N(1)}, 0\} = \max\{\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}, 0\} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

and set $x_{N(1)} = \min \frac{(A_B^{-1} b)}{d} = \min\{\frac{2}{2}, \infty\} = 1$. To find out which index we should remove from B, we compute

$$x_B = A_B^{-1}(b - A_N x_N) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

This implies that we need to replace $B(1)$ with $N(1)$. The new index set is $B = \{1, 3\}$, and

$$A_B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}, \quad A_N = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad c_B^T = (3, 0), c_N^T = (2, 0).$$

We observe $A_B^{-1} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$ and the new "modified objective"

$$c_N^T - c_B^T A_B^{-1} A_N = (2, 0) - (3, 0) \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}(1, 3),$$

Hence we have found the minimum. To determine its coordinates, we simply compute

$$x_B = A_B^{-1}(b - A_N x_N) = A_b^{-1} b = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

The optimal solution is $x^* = \begin{pmatrix} 1 \\ 0 \\ 3 \\ 0 \end{pmatrix}$.

(The minimum value $3 = \min f(x^*)$.)

2. The objective function is already linear. The constraint $||x||_1^2 \leqslant 9$ is equivalent to $|x_1| + |x_2| \leqslant 3$. From the lectures we know that this constraint can be modelled by

$$|x_1| \leqslant z_1, |x_2| \leqslant z_2, z_1 + z_2 = 3, z_1, z_2 \geqslant 0$$

and hence

$$\max x_1 + x_2 \text{ subject to } \quad -z_1 \leqslant x_1 \leqslant z_1, -z_2 \leqslant x_2 \leqslant z_2, z_1 + z_2 = 3, z_1, z_2 \geqslant 0.$$

There are alternative models, but since they have not been introduced in lectures, detailed explanation must be given.

Problem 3 (5 points). Consider the payoff matrix:

	$B1$	$B2$	$B3$
$A1$	2	-1	3
$A2$	0	4	-2

1. Determine if any strategies are dominated. Show your work.
2. Find the optimal mixed strategy for Player A using linear programming.

Solution 3. 1. Since $2 > 0, -1 < 4$, neither strategy A1 nor A2 dominates the other.

Since $2 > -1, 0 < 4$ neither strategy B1 nor B2 dominates the other. Similarly, since $-1 < 3, 4 > -2$ neither strategy B2 nor B3 dominates the other. Finally, since $1 < 3, 0 > -2$ neither strategy B1 nor B3 dominates the other.

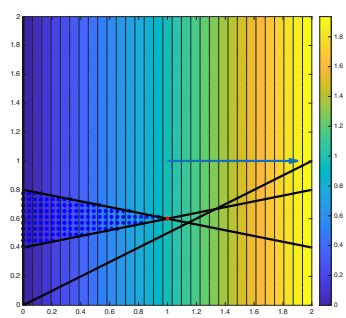
2. Let $A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & -2 \end{pmatrix}$ be the payoff matrix. From lectures we know that the optimal mixed strategy is given by the linear programming problem

$$\begin{aligned} \max v \quad \text{subject to} \quad A^T x = \begin{pmatrix} 2 & 0 \\ -1 & 4 \\ 3 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &\geqslant \begin{pmatrix} v \\ v \\ v \end{pmatrix} \\ (1,1) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= x_1 + x_2 = 1 \\ x &\geqslant 0 \\ v &\in \mathbb{R} \end{aligned}$$

which we can reformulate using $x_2 = 1 - x_1$ as

$$\begin{aligned} \max v \quad \text{subject to} \quad v - 2x_1 &\leqslant 0 \\ v + 5x_1 &\leqslant 4 \\ v - 5x_1 &\leqslant -2 \\ x_1 &\geqslant 0 \\ v &\in \mathbb{R} \end{aligned}$$

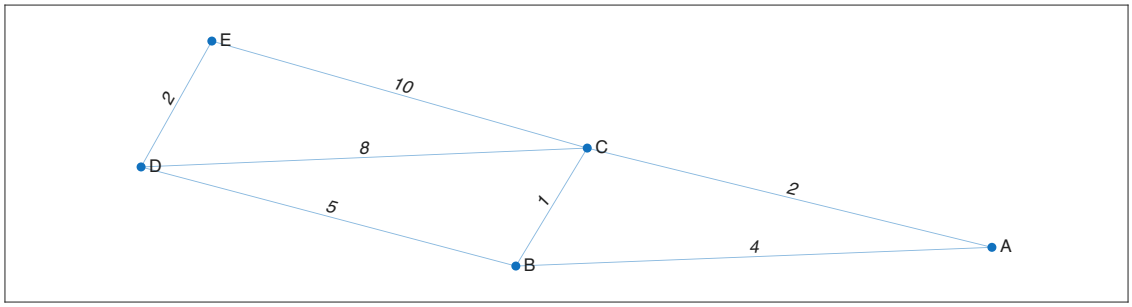
By the graphical method



we obtain $x_1 = \frac{3}{5}, x_2 = 1 - \frac{3}{5} = \frac{2}{5}$ with optimal payoff $v = 1$.

Problem 4 (5 points).

1. Find the shortest path from node A to node E using Dijkstra's algorithm for the following network:

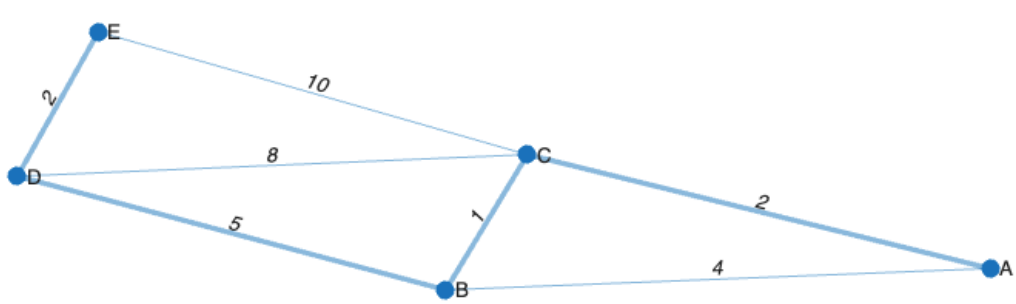


2. Provide the shortest path tree.

Solution 4. 1. Let $d = (0, \infty, \ldots, \infty)^T, p = (\emptyset, 0, \ldots, 0)^T$, and$v = (0, \ldots, 0)^T$, denote the distance, previous-node, and visited-node vectors, respectively. At each iteration, we pick a not-yet-visited node with shortest distance, mark it as visited, and update the distance of its neighbors if passing through this node is a shorter path. Here is the evolution of these vectors as the algorithm proceeds

$$
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 0 \\
B=2 & \infty & 0 & 0 \\
C=3 & \infty & 0 & 0 \\
C=4 & \infty & 0 & 0 \\
E=5 & \infty & 0 & 0
\end{array}
\longrightarrow
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 1 \\
B=2 & 4 & 1 & 0 \\
C=3 & 2 & 1 & 0 \\
D=4 & \infty & 0 & 0 \\
E=5 & \infty & 0 & 0
\end{array}
\longrightarrow
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 1 \\
B=2 & 3 & 3 & 0 \\
C=3 & 2 & 1 & 1 \\
D=4 & 10 & 3 & 0 \\
E=5 & 12 & 3 & 0
\end{array}
\longrightarrow
$$

$$
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 1 \\
B=2 & 3 & 3 & 1 \\
C=3 & 2 & 1 & 1 \\
D=4 & 8 & 2 & 0 \\
E=5 & 12 & 3 & 0
\end{array}
\longrightarrow
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 1 \\
B=2 & 3 & 3 & 1 \\
C=3 & 2 & 1 & 1 \\
D=4 & 8 & 2 & 1 \\
E=5 & 10 & 4 & 0
\end{array}
\longrightarrow
\begin{array}{lccc}
 & d & p & v \\
A=1 & 0 & & 1 \\
B=2 & 3 & 3 & 1 \\
C=3 & 2 & 1 & 1 \\
D=4 & 8 & 2 & 1 \\
E=5 & 10 & 4 & 1
\end{array}
$$

The shortest path from A to E is $A = 1 \to C = 3 \to B = 2 \to D = 4 \to E = 5$, the cost is 10.

2. The shortest path tree is